

Revision Report / Response to Reviewers

Initial Placement for Fruchterman–Reingold Force Model With Coordinate Newton Direction

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Stephen Kobourov
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Dear Prof. Stephen Kobourov,

Thank you for directing us to the reviewers’ comments and for giving us the opportunity to revise our manuscript. We confirm that we have received the review reports.

Based on the reports, we have carefully reviewed all comments and revised the manuscript accordingly. Our responses are provided below in a point-by-point manner. Where applicable, we refer to section, equation, figure, or algorithm numbers in the revised manuscript.

We hope the revised version is now suitable for publication and look forward to hearing from you.

Sincerely,

Hiroki Hamaguchi, Naoki Marumo, and Akiko Takeda

A Response to Reviewer A

We thank Reviewer A for the careful and constructive assessment of our manuscript. We appreciate the reviewer’s observation that the proposed algorithm is interesting and that there are no major problems with the construction of the algorithm. We have revised the manuscript to address the concerns regarding the relationship between the proposed method and the Fruchterman–Reingold model, as well as the distinction between the FR and KK models.

A.1 Relationship Between the Proposed Approach and the FR Force Model

First, the authors do not sufficiently consider how well their approach matches the FR force model.

Originally, Fruchterman and Reingold [10] proposed the FR model and solved it by using a pseudo-physical simulation-based algorithm (by following Eades’ spring embedder method [8]). By contrast, Kamada and Kawai [21] proposed the KK force model and solved it by a Newton-like method. Also, many succeeding studies followed these approaches, i.e., either by combining simulation-based algorithms with the FR model or by combining Newton-like methods with the KK model. Although Hosobe [19] showed that both models could be solved with a Newton-like method called L-BFGS, the results still suggested that the KK model was more suitable for L-BFGS than the FR model.

However, this submitted paper adopts an unusual approach that combines a Newton-like method with the FR model. I understand that the paper is focused only on the

computation of initial layouts, but I think that the validation of adopting this approach is still insufficient.

It should be noted that the experimental results in Section 5 show that the proposed initial layout algorithm works better when it is combined with the L-BFGS algorithm for computing final layouts. It seems from Figure 6 that the combination of the proposed initial layout algorithm with the FR algorithm for computing final layouts does not work well except in limited cases like jagmesh1.

If the authors think that the proposed algorithm should be combined with the L-BFGS, they should adopt the KK model instead of the FR model, which would be more natural and more persuasive. If the authors think that the proposed algorithm works well for the FR layout algorithm (or another simulation-based algorithm), they should more deeply explore this direction and present clearer results in the paper.

Response: ご指摘に感謝いたします。原稿では確かなになぜ我々が FR モデルに焦点を当てたのかを十分に正当化できていませんでした。

重要なのは次の2点です。第一に、我々はやはり FR モデルを主対象とすることには意味があると考えています。それが最も効果のある問題設定であると考えているからです。第二に、しかし提案手法が KK モデルなどの他の graph-induced objective functions にも親和性があることは仰る通りで、そこに説明の余地があります。そこで、提案手法のより一般的な形として類似した座標降下型の最適化手法も提案し、提案手法はその特殊なケースとして位置づけます。そして、一般的な手法においても同様の数値実験も追加で行いました。

まず、第一の点として、なぜ FR モデルを主対象とするのかについて説明します。

我々が FR モデルに焦点を当てた理由は主に二つあります。第一に、FR 目的関数には、提案する座標ごとの計算によって活用できる edge-based な疎構造が含まれているためです。提案手法ではこの疎性を座標更新に自然に反映できます。この点は、距離保存項が本質的に頂点对上に定義される KK stress model とは異なります。したがって、FR モデルは、提案手法の sparsity-based advantage を明確に示す上で適した対象であると考えています。ただし、KK モデルでは目的関数が通常、頂点对に関する pairwise terms を含むため、FR 目的関数の edge-based attractive term に対して利用できるような疎性は大部分失われます。したがって、我々が FR モデルに焦点を当てたのは、提案する枠組みが KK に適用できないことを意味するものではなく、提案手法の計算上の利点が最も明確に現れる設定を主対象としたためです。

第二に、本研究の出発点の一つは、FR モデルを連続最適化問題として定式化した場合、L-BFGS のような手法によって、古典的な simulation-based FR algorithm と比べて最終レイアウト品質を大きく改善できる余地があるという点です。査読者のご指摘された通り、我々の実験でも、提案する初期配置は、古典的な FR algorithm と組み合わせた場合よりも、L-BFGS と組み合わせた場合により明確な効果を示しています。これは本論文の主張と矛盾するものではなく、むしろ FR objective を連続最適化の観点から扱う場合に、良い初期配置が重要になることを示していると考えています。

よって、これらの理由を明確に記すことによって、ご懸念の点に対して説明を果たす所存です。

次に、第二の点として、提案手法が KK モデルや他の graph-induced objective functions にも適用できることを説明する新しい section を追加しました。ここでは、提案する座標ごとの手法を KK モデルに対してどのように導出できるかを説明し、同様の数値実験も行いました。

効率は問題の構造上やや落ちるものの、同様の傾向自体は確認できます。そのことを明記することにより、本論文がより広い示唆を含むものになると考えています。

A.2 Difference Between FR and KK Models

Second, the authors do not sufficiently consider differences between the FR force model and others, especially the KK model, which becomes apparent especially in Subsection 7.1.

Here the authors first say “Although we utilized the coordinate Newton direction only for the optimization problem (1), we can see that its application is not necessarily limited to the FR force model alone”, and then present “objective functions arising from graphs”. Does the authors mean that this discussion applies to other force models including the KK model? If so, I think that the authors should adopt the KK model instead of the FR model.

In addition, the authors present the graph isomorphism problem as an application of the proposed algorithm. I think that the KK model would be more suitable for this application than the FR model.

The following might work as hints for the authors’ paper revision. Intuitively, the FR model exhibits soft behavior while the KK model exhibits hard behavior. Therefore, the FR model is usually used for dynamic graph layouts like animation and interaction while the KK model is usually used for static layouts. Also, the FR model is sometimes used for very large graphs because simulation-based algorithms for the FR model can be more scalable than Newton-like algorithms for the KK model. One plausible direction for the authors might be to combine the proposed algorithm with a scalable simulation-based algorithm for the FR model and to apply it to very large graphs.

Response: 示唆に富むご指摘に感謝いたします。大まかには、First の点で述べたことと重なる部分もありますが、一般化は可能であり、それを基に議論しました。

graph isomorphism への応用に関するご指摘にも同意いたします。この応用は FR モデルに特に自然なものとは言い難く、本論文の主たる貢献から読者の注意を逸らす可能性があるため、改訂稿ではこの議論を削除します。

一方で、FR モデルの soft behavior に関する査読者のご指摘は、本研究の位置づけを明確化する上で非常に有益であると考えています。改訂稿では、提案手法の重要な応用として、dynamic または animation-like なグラフ描画シナリオに関する議論を強化します。具体的には、動的グラフにおける各時刻のレイアウト更新を、変更後の FR objective に対する再最適化のための初期配置問題として捉えます。このような設定では、すでに安定したレイアウトが得られており、そこに少数の頂点や辺が追加される場合、あるいはレイアウトやグラフ構造に小さな摂動が加えられる場合に、提案する座標ごとの手法によって、改善された初期レイアウトを効率的に計算できます。この利用場面は FR モデルの soft behavior とよく整合しており、また査読者が示唆された scalability-oriented な方向性とも整合しています。したがって、この側面を示すために、大規模かつ animation-like なグラフ描画シナリオに関する新しい実験を追加します。

A.3 Notation

The following are minor comments.

- Symbol E is used for two different purposes, one to indicate a set of graph edges and the other to indicate energies between nodes possibly with subscripts and superscripts. It will be better to use different symbols for these different concepts.
- I wonder if $(i, j) \in V^2$ with $i \neq j$ in (1) is appropriately intended. This treats both (i, j) and (j, i) in the sum.
- Symbol ϵ appears only once in Subsection 4.4, which seems strange.

- Symbol f_i at line 6 of Algorithm 1 probably lacks the superscript “a”.
- In Subsection 5.3, the word “L-BFGS” that appears as superscripts of N does not seem to be correctly specified. The hyphen mark looks too long.
- In the optimization problem called “objective functions arising from graphs” in Subsection 7.1, “=” probably should be “:=”.
- The header of Subsection 8.1 seems to have a problem with the hyphen mark again.

Response: Thank you for pointing out these notation and typography issues. We addressed them as follows.

- We avoided the symbol conflict by using a dedicated symbol for the edge set (so that the energy terms $U_{i,j}$ are not confused with the edge set).
- We intended the sum to be over all ordered pairs of distinct vertices, so that both (i, j) and (j, i) are included. We clarified this in the text and notation.
- We clarified the role of ϵ as the minimum separation in the discrete point set Q and its use in preventing divergence.
- We corrected the missing superscript in Algorithm 1 (the update uses the attractive-only objective f_i^a).
- Regarding the reviewer’s note on the “L-BFGS” superscripts: the hyphen was already intended as a proper hyphen in math mode, but its appearance can still be misleading. To avoid confusion, we now use the spelling “LBFGS” when it appears inside math expressions (e.g., in superscripts), while keeping the standard spelling “L-BFGS” in running text.
- We corrected the definition symbol in the “objective functions arising from graphs” formulation.
- We checked the subsection headers and ensured hyphenation uses proper LaTeX conventions.

B Response to Reviewer B

We thank Reviewer B for the careful review and constructive suggestions. We appreciate the positive assessment that the paper addresses a topic of clear interest to the JGAA audience, that the technical explanation is comprehensive, and that the mathematical treatment is mostly easy to follow. We have revised the manuscript to improve the framing of the contribution, the discussion of related work, the experimental evaluation, and the organization of the later sections.

B.1 Related Work on Multilevel Approaches

In the Related Work section, in 3.1 Multilevel approaches are quickly dismissed. I don’t understand the statement the statement about `sfdp` being “prior work” to FR - as the latter came way later than the former. This sentence should be clarified. Also, I believe that further space could be given to multi-level approaches, also considering that could be one potential application/extension of this technique beyond this paper.

Response: Thank you for pointing this out. We clarified the statement around `sfdp`: our intent was to reference it as prior work on scalable, multilevel force-directed graph drawing rather than implying chronological priority over the FR model. We also expanded the discussion in the related-work section and clarified how our method is complementary to multilevel pipelines (in particular, as an improvement to the refinement stage).

Multilevel approaches such as `sfdp` in Graphviz are important prior work on scalable force-directed graph drawing. These methods use hierarchical coarsening and refinement to achieve efficient layout computation for large graphs.

Our work is complementary to such multilevel methods. Optimization techniques such as L-BFGS can be directly applied to the refinement stages within multilevel pipelines. Since the approaches proposed in this study can be incorporated into multilevel frameworks with appropriate modifications, our primary objective is to quantify and enhance the refinement procedure itself. Therefore, we deliberately exclude coarsening and other approximation steps from the present analysis.

B.2 Uncertain Framing of FR Versus L-BFGS

The framing of this paper is somewhat uncertain. Despite the paper title and general orientation towards FR, L-BFGS is consistently and repeatedly mentioned as a better alternative. Even in the related work section, in Section 3.1, only L-BFGS is mentioned as a ready-to-go measure for application in Multilevel approaches. In the experiments the paper explicitly mentions that FR is consistently outperformed by L-BFGS - then why we should use FR (even with the improved initial placement) at all? The paper does not make a good argument why we should choose that over the other, thus in a way making me question even the title of the paper. In general, it's unclear what's the impact of this research and how researchers could build upon it.

Response: FR を使うべきと思われる状況は、例えば、動的グラフのレイアウト更新のようなシナリオであると考えています。動的グラフでは、各時刻のレイアウト更新を、変更後の FR objective に対する再最適化のための初期配置問題として捉えることができます。reviewerA でも指摘されたように、FR モデルの soft behavior はこのようなシナリオとよく整合しており、また reviewerB が示唆された scalability-oriented な方向性とも整合しています。したがって、この側面を示すために、大規模かつ animation-like なグラフ描画シナリオに関する新しい実験を追加します。

B.3 Expansion of the Evaluation Dataset

The evaluation yields positive results, but in my opinion it could be further improved. First, I would suggest to expand 5.2 it in terms of number of graphs. More graphs from the Florida library could be used, or another source could be [1]. This expanded datasets already includes 3elt, but expands also on grids, cylinders, and other regular graphs that make up a solid benchmark set, and will provide a wider set of instances to discuss. If not all figures can be included in the main body of the paper, these could be moved in the supplemental material.

[1] Bartel, Gereon, et al. "An experimental evaluation of multilevel layout methods." International Symposium on Graph Drawing. Berlin, Heidelberg: Springer Berlin Heidelberg, 2010.

Response: Thank you for the suggestion. 実験を追加します。(Appendix などで良さそう)

B.4 Visualization and Color Map

Concerning the visualization of the layout, I don't believe that the index colouring adds much. A rainbow map, however, it's not a great choice. If color should be kept, maybe I would consider using a sequential continuous color map with a legend (see [2] for examples).

[2] <https://matplotlib.org/stable/users/explain/colors/colormaps.html>

Response: Thank you for the suggestion. We agree that the current index-based coloring does not add much interpretive value. We revise the visualization by replacing the rainbow map with a sequential continuous color map (Since some graphs such as jagmesh1 have a symmetric structure, the color can help visually distinguish different vertices). We also clarified the color only represents the vertex index for visual distinction.

B.5 Effect of Initial Placement Over Time

One thing that comes out of the charts in Figure 6 is that this different initial positioning has limited effect on the way the objective function decreases over time - which is an interesting phenomenon. In FR, the chart appears shifted below by how "better" the initial positioning is, with the non-CN version never catching up. Differently, it seems that the objective functions of L-BFGS and CN-L-BFGS tend to converge before the iteration cap. Does this mean that the positive effect of the initial placement tends to fade faster with L-BFGS, despite the better initial condition?

Response: Yes, this interpretation is consistent with our observations. We added a brief explanation in the experimental discussion: for FR, a better initialization tends to yield a persistent offset in the objective curve, whereas for L-BFGS the curves often converge earlier so the relative advantage can diminish once near-stationary points are reached. We also clarified that the effect should be judged not only by the final objective value, but also by the early-stage convergence behavior and runtime.

B.6 Singularities in dwt_992 and collins_15NN

The dwt_992 and collins 15NN singularities could be described more. Would it be possible to generalize, highlighting what makes these two cases so different from each other?

Speaking of singularities, in the charts it seems like there is a spike in $f(X)$ right after the 0 seconds mark (like in 3elt, btree or cycle) is that an artifact?

Response: todo?

In some FR runs, a small increase in $f(X)$ appears immediately after the start of the iterations; this can happen because the first few displacement steps are relatively large before the temperature decreases, so it is an artifact of the early-stage update schedule rather than a failure of the initialization.

The effect of the initial placement also differs across the two solvers: for FR, a better initialization tends to produce a persistent offset in the objective curve, whereas for L-BFGS the curves often converge earlier so the relative advantage can fade near stationary

points. Thus, the benefit of initialization should be judged not only by the final objective value, but also by the early-stage convergence behavior and runtime.

B.7 Organization of Sections 6 and 7

I am less persuaded from Sections 6 and 7. Section 6 includes considerations that I would include in Section 4, as to further justify your optimization design. I would also simplify the discussion, removing subsections and integrating their contents in Section 4. In Section 7 I would focus on the impact of this approach on graph drawing more specifically. For example, where is the next step for this technique, in the context of graph drawing? Will its presence in a drawing pipeline make a difference, or the gains in quality are not sufficient to justify its use? What about its inclusion in a multi-level pipeline (see also a similar point above)?

Response: これは応用例の section を ReviwerA で答えたのと被る。応用例を消すか、増やすか、記述をもっと精緻化するか。

B.8 Moving the Section 8 to the Appendix

I would move Section 8 to supplemental material out of the main body of the paper.

Response: Thank you. In the revision, we moved the implementation-oriented material (NetworkX implementation details) to the Appendix so that the main narrative focuses on the method, experiments, and discussion.

B.9 Bibliography

Bibliography must be updated. Some papers do not include any venue, others do not have a DOI, others only report their archive version (while journal versions are available).

Response: これが断トツで自信がないが、対応が不完全。Thank you for the careful note. In the revision, we reviewed the bibliography entries and improved completeness where possible (e.g., adding missing metadata and cleaning local-only fields), and we checked for cases where a journal/conference version should be cited instead of an archive-only version.

Closing

We again thank the editor and the reviewers for their valuable comments. We believe that the revised manuscript is clearer in scope, better positioned with respect to existing force-directed graph drawing methods, and more explicit about both the strengths and limitations of the proposed approach.

Sincerely,

Hiroki Hamaguchi, Naoki Marumo, and Akiko Takeda